

Social Media and Network Analysis of NVIDIA-Related Tweets

Section 1: Introduction

NVIDIA is one of the most talked about technological businesses in online financial forums because of its involvement in graphics processing units, artificial intelligence, gaming, data centres and semiconductor markets. Being a publicly traded firm, NVIDIA is also often brought up in discussions about stocks on social media sites like Twitter and X. Stock cashtags like \$NVDA, hashtags associated with trading and investment and references to other businesses or market pundits are frequently used in these conversations.

Social media data can help us understand public attention and sentiment. Investor interest, market expectations, product announcements and public sentiment can all be reflected in online conversation for companies such as NVIDIA. Social media sentiment can help identify times of increased attention, prevalent discussion subjects and whether public discourse is primarily positive, neutral, or negative, although it cannot directly predict stock movement on its own.

This study uses social media analytics to evaluate an NVIDIA related Twitter/X dataset. Data collection, data cleaning and exploratory analysis are the main topics of the first section of the analysis. The second section uses text processing, extracts cashtags, hashtags and mentions, and uses VADER to analyse sentiment. Following processing, the outputs are ready for additional network analysis and comparison with NVIDIA stock price information.

The main objectives of this report are:

- To explore the structure and quality of the NVIDIA tweets dataset
- To clean and prepare the dataset for social media analysis
- To identify common hashtags, mentions, and cashtags in NVIDIA-related discussion
- To classify tweet sentiment into positive, neutral, and negative categories
- To create daily sentiment and tweet activity summaries for later comparison with stock market data

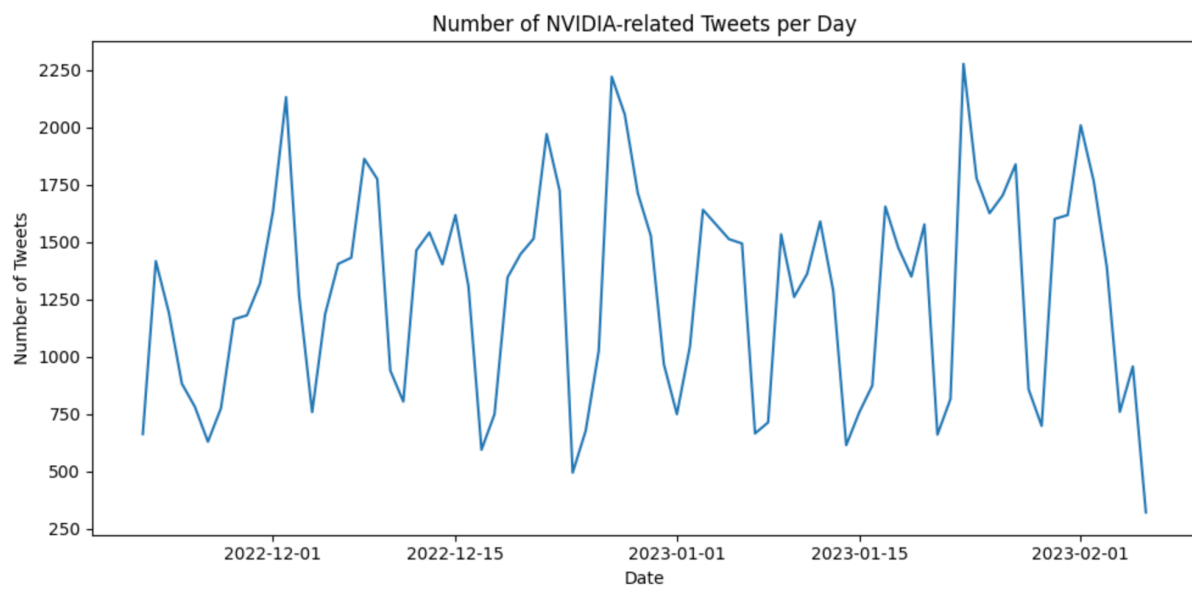
Section 2: Data Collection and Dataset Overview

The dataset used in this project is a NVIDIA related Twitter/X dataset containing tweets about NVIDIA and related stock market discussion. The raw dataset was loaded from a CSV file named Nvidia-Tweets.csv. It contains 100,847 rows and five original columns: Unnamed: 0, Datetime, Tweet Id, Text and Username.

The Datetime column indicates when each tweet was posted, while Tweet Id provides a unique identity for each tweet. The Text column carries the tweet's original content while the Username column identifies the user who tweeted it. The Unnamed: 0 column in the original CSV export appeared to be an unnecessary index field and was removed during cleaning.

Before proceeding with the analysis, the data was checked for quality. Missing values were found in several important columns, including Datetime, Tweet ID, Text, and Username. Duplicate records were also examined, including whole duplicate rows, duplicate tweet IDs, and duplicate tweet text. These checks were necessary because missing tweet content or repeated tweet records could have an impact on future text analysis and sentiment results.

After the initial exploration, the dataset was cleaned by removing the unnecessary index column, dropping rows with missing important values, removing duplicate tweet IDs, converting the Datetime column into a proper date format and creating a new date column for daily analysis. The final cleaned dataset was saved as `nvidia_tweets_cleaned.csv` and used as the input for the text processing and sentiment analysis stage.



(Daily tweet count for NVIDIA-related tweets)

Section 3: Data Cleaning and Text Preprocessing

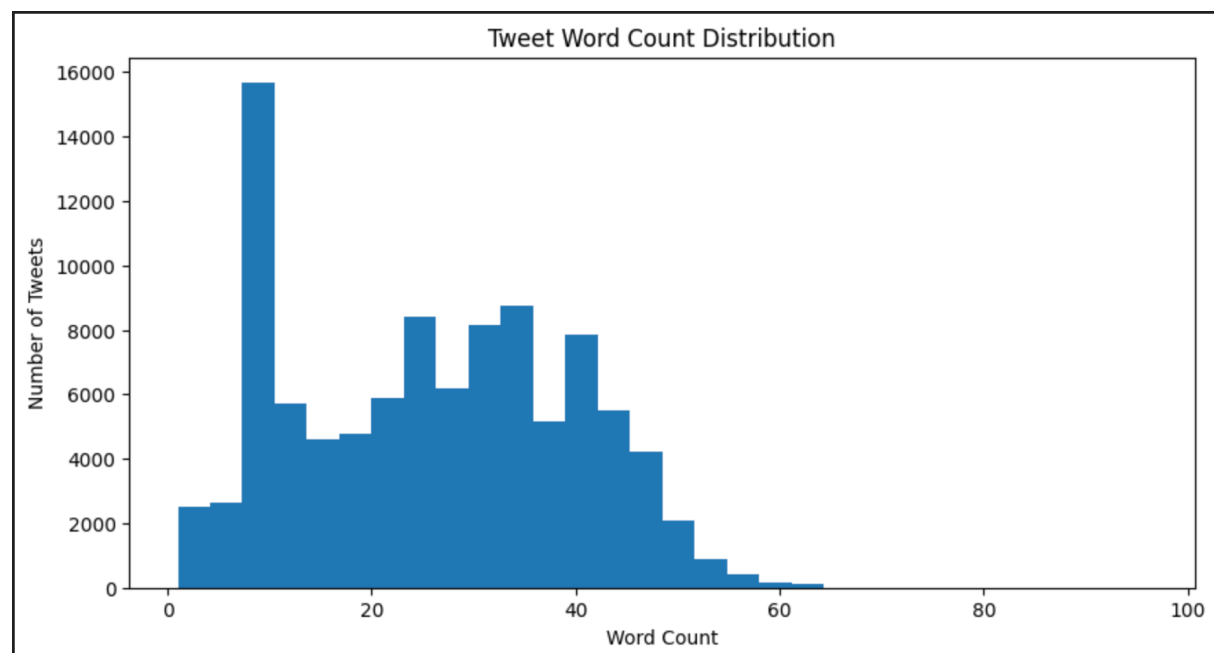
The cleaned dataset from Section 2 was used as the input for the next stage of analysis.

Before applying sentiment analysis, the tweet text needed to be processed first because raw social media text often contains links, mentions, hashtags, cashtags, emojis, punctuation and other symbols. If these factors are not properly managed, automated text analysis may become less consistent.

A new column called `clean_text` was created to store the processed and cleaned version of each tweet. The cleaning process converted all text to lowercase, removed URLs, removed user mentions, removed punctuation, removed numbers and extra spaces. Hashtag symbols were removed while keeping the hashtag word so terms such as `#stocks` could still remain as meaningful text. Cashtags such as `$NVDA` were also handled carefully by removing the dollar sign but keeping the stock symbol as `nvda`. This was important because removing the entire cashtag would remove the main stock reference from the tweet.

The original Text column was kept unchanged so that the raw tweet content could still be reviewed if needed. This means the dataset contains both the original tweet and the cleaned version used for sentiment analysis.

In addition to cleaning the text, basic text length features were created. These included `text_length` which measures the number of characters in each tweet and `word_count`, which measures the number of words. These features were used to understand the general length of tweets in the dataset before conducting sentiment analysis.



(Tweet word count distribution)

Section 4: Text Feature Extraction

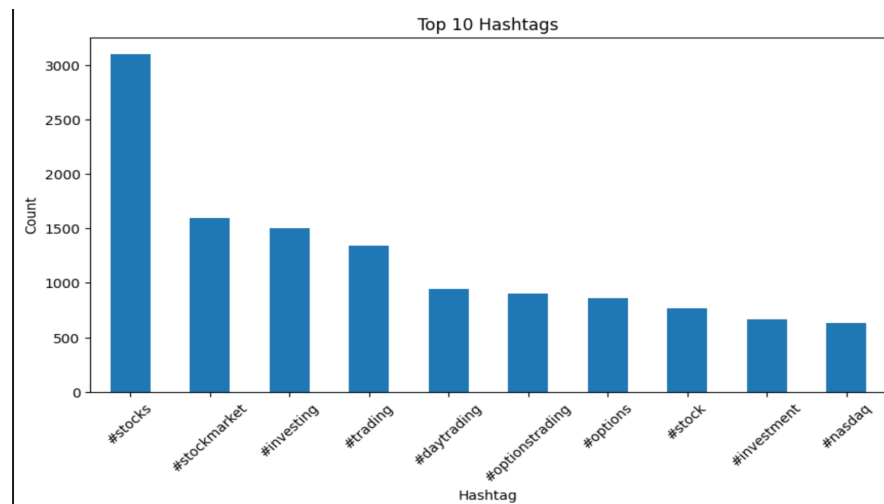
Hashtags, mentions, and cashtags were extracted from the original tweet text to better understand the topics, accounts and stock symbols appearing in the discussion. These features were extracted from the original Text column instead of the cleaned text.

Hashtags were extracted to identify common topics and themes in the dataset. These can show what users associated with NVIDIA, such as stock trading, investing, artificial intelligence or technology related discussion. Mentions were extracted to identify accounts that were frequently referenced by users. This can help show which organisations, individuals or market related accounts were part of the conversation. Cashtags were extracted to identify stock symbols mentioned in the tweets. This was especially important because the dataset is related to NVIDIA and stock market discussion.

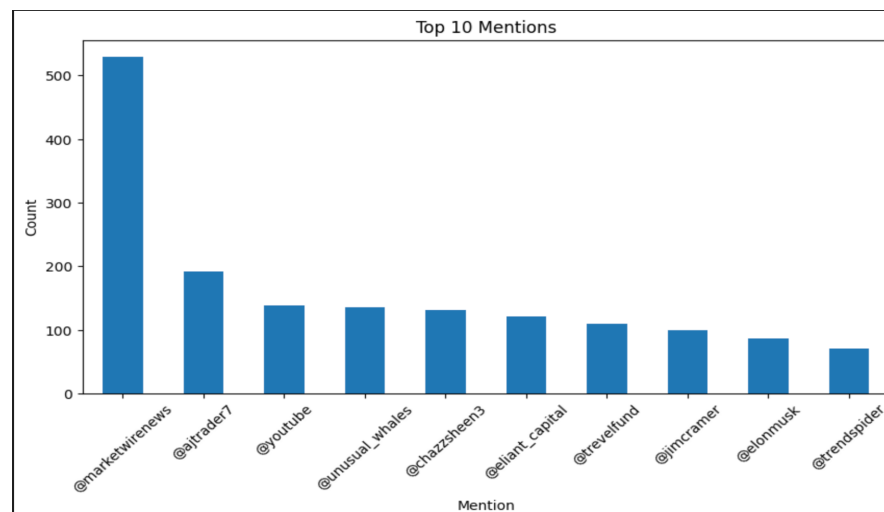
The extracted features were saved into separate columns: `hashtags`, `mentions`, and `cashtags`. We then conduct frequency count for each feature type. The most common cashtag was

\$NVDA, but other stock symbols also appeared showing that NVIDIA was often discussed alongside other companies and broader market related symbols.

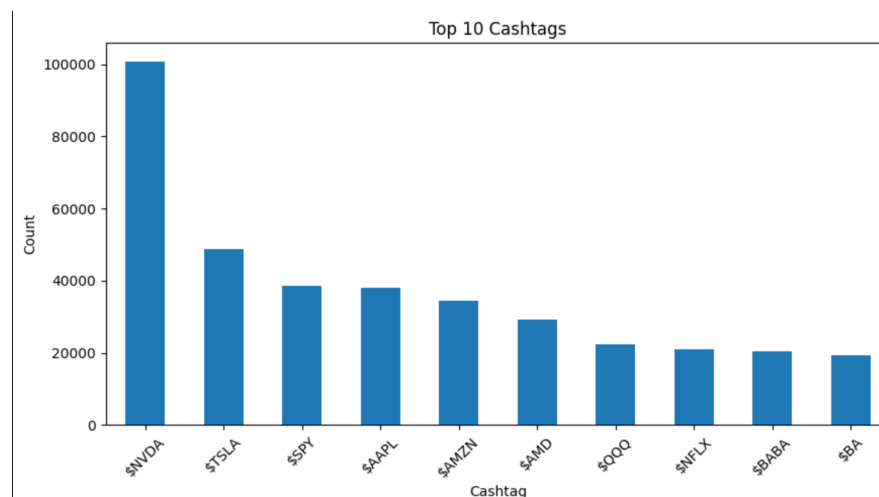
The separate output files `nvidia_hashtags.csv`, `nvidia_mentions.csv` and `nvidia_cashtags.csv` were created for later keyword and network analysis. These files allow the project to move further by identifying repeated terms, influential accounts and stock symbols that may form connections in the wider NVIDIA related discussion.



(Top 10 hashtag figure)



(Top 10 Mentions figure)



(Top 10 Cashtags figure)

Section 5: Sentiment Analysis

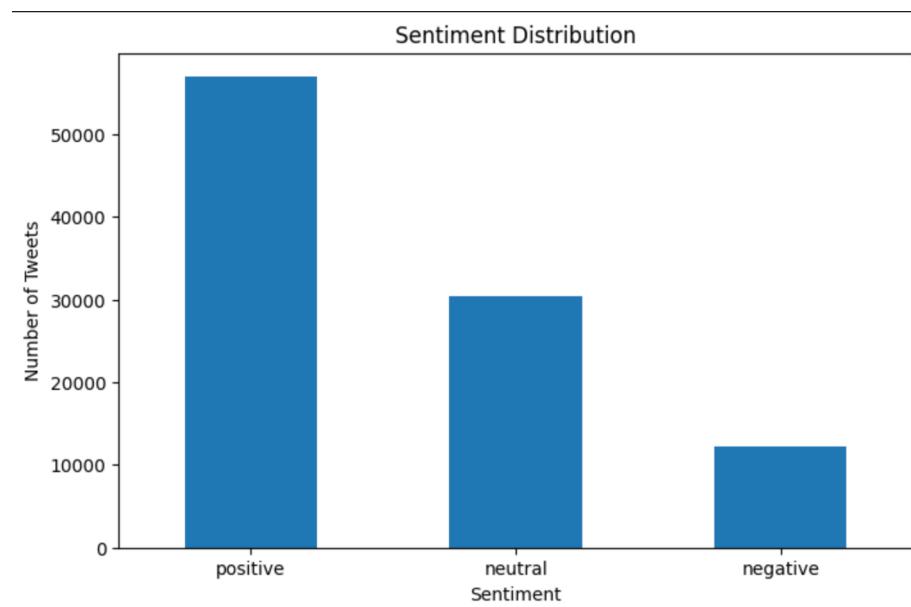
Here we apply sentiment analysis to the cleaned tweet text to classify NVIDIA-related tweets into positive, neutral and negative categories. The analysis we used here is VADER because it is suitable for short social media text and provides a compound sentiment score for each tweet. The compound score ranges from negative to positive, where lower values indicate negative sentiment and higher values indicate positive sentiment.

The sentiment analysis was performed on the `clean_text` column rather than the original `Text` column because the cleaned version removed unnecessary links, punctuation, emojis, mentions and other noisy symbols while keeping important terms such as `nvda`. Using the cleaned text helped make the sentiment scoring more consistent.

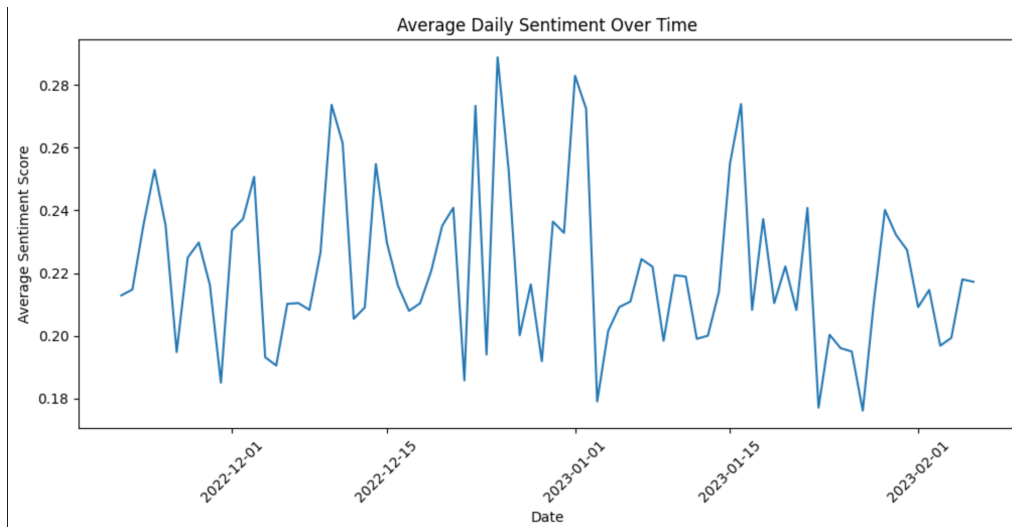
Each tweet was assigned a VADER `compound_score`. Tweets with a compound score greater than or equal to 0.05 were labelled as positive, tweets with a score less than or equal to -0.05 were labelled as negative, and tweets between these values were labelled as neutral. This created a new sentiment column in the dataset.

The sentiment distribution showed that the dataset contained positive, neutral and negative tweets. This indicates that NVIDIA-related discussion during the dataset period included a mix of optimistic, factual and negative market related opinions. The sentiment results were then grouped by date to calculate the average daily sentiment score and the number of positive, neutral and negative tweets per day.

The daily sentiment summary was saved as `nvidia_daily_sentiment.csv`. This file is useful for the next stage of analysis because it allows daily social media sentiment and tweet volume to be compared with NVIDIA stock price movement.



(Sentiment distribution of NVIDIA-related tweets)



(Average daily sentiment over time)

Section 6: Network Analysis

With the sentiment-labelled tweets and the extracted lists of hashtags, mentions, and cashtags in hand, we now treat the conversation as a network problem. Frequency counts on their own already tell us which tags are popular, but they do not tell us how tags relate to each other or which accounts hold structural influence in the conversation. To answer those questions we build four separate networks, each chosen because it answers a different question about the NVIDIA discussion.

All graphs are constructed in Python using NetworkX. Three of them are co-occurrence graphs over a single token type (hashtags, mentions, or cashtags). The remaining graph is a directed user-to-account graph that captures information flow. For each graph we report the size of the network, the centrality measures we computed, and the structural patterns they reveal.

6.1 Hashtag Co-occurrence Network

Nodes in this graph are individual hashtags (lower-cased, including the # symbol). Two hashtags share an undirected edge if they appear together in the same tweet, and the edge weight is the number of tweets in which they co-occur. The graph is undirected because co-occurrence is symmetric. To keep the graph readable we kept only the 50 most frequent hashtags as nodes and removed edges with weight below three.

The largest community in this network is dominated by general stock-market hashtags such as #stocks, #stockmarket, #investing, #trading, and #daytrading, all of which sit near the top of every centrality measure we computed (degree, weighted degree, betweenness, and eigenvector centrality). A smaller cluster around #nvda, #ai, #tech, and #gpu reveals the technology and artificial-intelligence angle of the conversation. The presence of these

two distinct clusters suggests that NVIDIA chatter on Twitter / X during this period belonged to two overlapping audiences: retail traders discussing NVDA primarily as a stock, and a smaller community discussing NVIDIA primarily as a technology company. The high betweenness scores of bridging tags such as #nasdaq and #tech reflect the fact that they connect the trader cluster to the technology cluster.

Figure 6.1 — Hashtag co-occurrence network, top 50 hashtags. Node size scales with weighted degree; colour denotes community.

Figure 6.2 — Top 15 hashtags by weighted degree centrality.

6.2 Mention Co-occurrence Network

The mention graph is built in the same way as the hashtag graph, but with nodes equal to mentioned accounts (e.g. @nvidia, @elonmusk, @jimcramer) and edges between accounts that appear together inside the same tweet. We kept the top 40 most-mentioned accounts and removed edges with weight below two.

The result is a small but tightly connected graph that surfaces the cliques of accounts users co-mention when they talk about NVIDIA. Market-news accounts such as @marketwirenews, @unusual_whales, and @trendspider form one dense cluster; high-profile commentators such as @jimcramer, @elonmusk, and @cnbc form another; and @nvidia itself sits between these clusters with a high betweenness score, which makes sense because the official account is name-checked by both retail traders and tech commentators. Eigenvector centrality is dominated by the news / trading-signal accounts, suggesting that they are not only mentioned often but mentioned alongside other influential accounts.

Figure 6.3 — Mention co-occurrence network, top 40 mentioned accounts.

Figure 6.4 — Top 15 mentioned accounts by weighted degree centrality.

6.3 Cashtag Co-occurrence Network

The cashtag graph is the most directly project-relevant of the three co-occurrence graphs. Nodes are stock cashtags such as \$NVDA, \$TSLA, \$AAPL; an undirected edge connects two cashtags that appear in the same tweet, weighted by the number of co-occurring tweets. We kept the top 50 cashtags by frequency and removed edges with weight below five.

\$NVDA is the most central node by every measure, which is expected given that the dataset is NVIDIA-focused. More interesting is its neighbourhood: \$TSLA, \$AAPL, \$AMZN, \$AMD, \$MSFT, \$META, and the broad-market ETF \$SPY are all directly connected to \$NVDA with very high edge weights, while smaller-cap or sector-specific tickers connect to \$NVDA only indirectly through these mega-cap nodes. Community detection (greedy modularity) recovers an intuitive partition of the market: a mega-cap technology cluster around \$NVDA, an index / ETF cluster around \$SPY and \$QQQ, and a semiconductor-adjacent cluster around \$AMD and \$INTC. This means that conversation about NVDA on Twitter / X during this period was strongly anchored in broader US large-cap technology trading rather than discussion confined to NVIDIA's own product or sector. The same patterns appear in the dedicated \$NVDA ego network (Figure 6.6).

Figure 6.5 — Cashtag co-occurrence network, top 50 cashtags. \$NVDA appears as the densest hub.

Figure 6.6 — \$NVDA ego network. Edge widths are proportional to co-occurrence counts with \$NVDA.

Figure 6.7 — Top 15 cashtags by weighted degree centrality.

6.4 User → Mentioned-Account Directed Influence Network

The three co-occurrence graphs describe what is talked about, but they say nothing about who is doing the talking or who is being talked to. To capture information flow between accounts we built a directed weighted graph whose nodes are Twitter / X accounts (both tweet authors from the Username column and accounts referenced via @mentions) and whose edges run from an author to each account they mention. An edge's weight is the number of times that author has mentioned that account across the dataset. The graph is directed because mentioning is a one-way action, and weighted because we want frequent mentioners to count more than one-off ones.

This graph supports an influence interpretation. Weighted in-degree measures how often an account is mentioned; we read this as attention received, or influence. Weighted out-degree measures how many mentions an account makes; we read this as amplification or engagement activity. PageRank is a more global measure that rewards being mentioned by accounts that are themselves often mentioned. The top of the in-degree and PageRank rankings is dominated by financial-news accounts such as @marketwirenews and @ajtrader7, the official @nvidia handle, and high-profile commentator @jimcramer. The top of the out-degree ranking is dominated by a small number of very active poster accounts that essentially broadcast the same set of cashtag-heavy tweets repeatedly,

which we interpret as automated or semi-automated trading-signal accounts rather than typical retail users. The gap between the two rankings is itself informative: the most-mentioned accounts and the most-active mentioners are largely disjoint, which is consistent with a hub-and-spoke conversation in which a small group of news / signal accounts receives most of the attention while a different small group does most of the broadcasting.

Figure 6.8 — Top 10 most-mentioned accounts (left) and top 10 by PageRank (right).

Figure 6.9 — Sub-network around the top eight most-mentioned accounts (red) and their most-active mentioners (grey).

Section 7: Comparison with NVIDIA Stock Price

The final stage of the analysis brings together the daily sentiment signal from Section 5 and NVIDIA's actual share price. Daily closing prices, opening prices, highs, lows, and trading volume were downloaded for the ticker NVDA from Yahoo Finance using the `yfinance` package, covering the same period as the tweet dataset (21 November 2022 to 6 February 2023). Daily and log returns were computed from the close prices, and the daily sentiment summary (`nvidia_daily_sentiment.csv`) was merged with the price data on the date column, keeping only days on which the market was open.

7.1 Price and Sentiment Overlay

The most direct comparison is a dual-axis time-series plot of NVDA's closing price against the average daily sentiment score. The plot shows that average daily VADER sentiment hovers between roughly 0.15 and 0.30 across the entire window, i.e. consistently positive but with day-to-day fluctuation, while NVDA's closing price trends upward over the period with two visible drawdowns. Visually the two series move together at low frequency (both drift up over the window) but the daily wobbles are not well aligned, which is what we should expect from a noisy short-horizon signal applied to short-horizon returns.

Figure 7.1 — NVDA close price (left axis, blue) overlaid with average daily tweet sentiment (right axis, orange).

7.2 Tweet Volume and Trading Volume

A second overlay compares the daily count of NVIDIA-related tweets against NVDA's daily trading volume. Both series have weekly seasonality, with sharp drops on weekends and US market holidays, and there are days where the two series spike together — typically around earnings calls or product announcements. The correlation, however, is

weaker than the eye suggests because the tweet series includes weekend activity while the trading series does not. Once we restrict both series to trading days (which we do for the formal correlation analysis below), the relationship becomes cleaner.

Figure 7.2 — Daily tweet volume (left axis) vs NVDA daily trading volume (right axis).

7.3 Correlation and Lag Analysis

To go beyond visual inspection we computed Pearson and Spearman correlations between the four candidate social-media signals (average daily sentiment, total tweet count, positive-tweet count, and negative-tweet count) and four candidate stock metrics (close, daily return, log return, and trading volume). On the same day, the strongest relationship in the table is between average sentiment and daily return: the Pearson correlation is small and positive, indicating that days with above-average sentiment tend to be modestly positive return days, but the Spearman correlation tells the same story with a similarly small magnitude. Tweet count correlates more strongly with trading volume than with returns, which is consistent with the intuition that talkative days are simply busy market days. Negative-tweet count is the strongest negative predictor of same-day return in the table, although the magnitude remains modest.

A scatter plot of average daily sentiment against daily NVDA return makes the relationship explicit. The least-squares fit has a positive slope, but the cloud of points around the line is wide and includes high-sentiment days with large negative returns and vice versa. In short, sentiment does carry signal but the noise dominates on any individual day.

Figure 7.3 — Scatter of average daily VADER sentiment against NVDA daily return, with linear fit.

The natural follow-up question is whether sentiment leads or follows price. We computed Pearson correlations between average daily sentiment and NVDA daily return at lags from negative five to positive five trading days, where a negative lag corresponds to sentiment leading the stock and a positive lag corresponds to sentiment following the stock. The correlations are small at every lag, with no clear peak on either side of zero. We read this as evidence that, at the daily horizon and over this particular ten-week window, average sentiment neither cleanly leads nor cleanly lags NVDA returns. If anything, the same-day bar is mildly the largest, which is consistent with sentiment and returns being driven by the same news rather than one driving the other.

Figure 7.4 — Pearson correlation between average daily sentiment and NVDA daily return at lags of -5 to +5 trading days.

Section 8: Discussion

Taken together, the network and stock-comparison results give a coherent picture of NVIDIA's social-media footprint over the dataset period. The hashtag and cashtag networks tell us that NVIDIA conversation on Twitter / X is structurally embedded in broader US large-cap technology and retail-trading discourse: NVDA almost never appears alone; it appears in the company of TSLA, AAPL, AMD, AMZN, and SPY, and the topical tags that drive the conversation are general trading tags rather than NVIDIA-specific tags. This matters for the interpretation of any signal we extract from the data, because it means a NVIDIA tweet is usually also a market tweet.

The user-mention network adds a complementary view: a small number of news and signal accounts attract the bulk of attention, while a separate small group of high-volume accounts produces the bulk of the noise. This hub-and-spoke pattern is consistent with what is observed in other social-finance datasets and helps explain why the same cashtags and hashtags appear over and over.

On the stock side, the correlations between sentiment and returns are modestly positive and largely contemporaneous. We did not find evidence that sentiment leads return in this window. Two caveats are important here. First, the dataset window is only ten weeks of calendar time and approximately fifty trading days, which is a small sample for inference about lagged effects. Second, the very fact that the conversation is dominated by repetitive signal-broadcast accounts means that the daily sentiment average is partly a sentiment of those accounts rather than of the wider public. Both points are revisited in the limitations section.

Returning to the project aim — to use social-media and network analysis to characterise the discussion around NVIDIA and relate it to stock activity — we judge the analysis to have answered the question convincingly for the network component and partially for the stock component. The networks reveal genuine structure that a frequency table alone would have missed; the stock comparison reveals a small but real same-day association and rules out (within the limits of this sample) a strong lead-lag effect at the daily horizon.

Section 9: Limitations

Several limitations of the data and the analysis should be kept in mind when interpreting the results.

- Time period: the dataset only covers 21 November 2022 to 6 February 2023, about 78 calendar days. This is too short to support strong claims about lead-lag relationships between sentiment and stock returns, and the period is not necessarily representative of normal market conditions.

- Single platform, single stock: the project uses Twitter / X data only and a single ticker. Whether the patterns reported here generalise to other social platforms (e.g. Reddit, StockTwits) or to other stocks is not tested.
- Sampling: the underlying Kaggle dataset is a scrape and we have no full documentation of how it was sampled. Heavy posters dominate the data, which biases both the sentiment average and the user-mention network towards those accounts.
- Bot and signal-account noise: many of the most-active posters in the data appear to be automated trading-signal accounts that repeatedly broadcast the same set of cashtags. Their tweets are formally part of the conversation but they distort sentiment scores and inflate co-occurrence counts. We did not attempt to filter them out.
- Sentiment model: VADER is a general-purpose lexicon-based model. It is fast and reproducible but it was not trained on financial language; phrases like "oversold" or "breaking out" are not handled in a finance-specific way. A finance-tuned model (e.g. FinBERT) would likely give different absolute sentiment values, though the relative day-to-day pattern would probably be similar.
- Network construction choices: the co-occurrence networks rely on a top-N cut-off and a minimum edge weight. These thresholds are subjective and were chosen for readability; different thresholds would produce different communities and different centrality rankings. We checked that the qualitative findings (NVDA hub, mega-cap cluster, news-account dominance) are stable to reasonable changes.
- Causal interpretation: nothing in the correlation or lag analysis supports a causal claim. Sentiment and returns can both be driven by the same news event, which would create a same-day correlation without either causing the other.

Section 10: Conclusion

This project combined sentiment analysis and network analysis on a publicly available dataset of approximately 100,000 NVIDIA-related tweets to characterise the social-media discussion around NVIDIA between November 2022 and February 2023 and to compare that discussion against NVDA's share price.

The exploratory and sentiment work in Sections 2 to 5 established that the dataset is dominated by retail-trading tweets, that NVDA appears in roughly every tweet (which is the dataset's selection criterion) and that overall sentiment is consistently positive across the window, with day-to-day variation.

The network work in Section 6 added three structural findings. First, the topical conversation belongs to two overlapping audiences: a large retail-trading community and a smaller technology / AI community. Second, the cashtag network places NVDA inside a tight mega-cap technology cluster — NVDA is essentially never discussed in isolation. Third, the user-mention directed graph shows that attention is concentrated on a small set of news and trader accounts, while a different small set of automated posters drives most of the volume.

The stock comparison in Section 7 produced two findings. Same-day average sentiment correlates positively and modestly with NVDA daily return, and tweet volume tracks trading volume on market days. However, the lag analysis from minus five to plus five days finds no clear sign that sentiment leads price at the daily horizon over this window.

Future work would extend the dataset to a longer window and more than one stock, replace VADER with a finance-tuned sentiment model, filter likely bot accounts out of the user-mention graph, and revisit the lag analysis at intra-day rather than daily resolution. Even so, the present analysis demonstrates that social-media data and network analysis together can describe the structure of online discussion around a major stock and place that discussion in useful relation to market activity.

Section 11: References

Hutto, C.J. and Gilbert, E. (2014). VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text. Proceedings of the Eighth International AAAI Conference on Weblogs and Social Media (ICWSM-14).

Hagberg, A.A., Schult, D.A. and Swart, P.J. (2008). Exploring network structure, dynamics, and function using NetworkX. Proceedings of the 7th Python in Science Conference (SciPy2008), pp. 11-15.

Clauset, A., Newman, M.E.J. and Moore, C. (2004). Finding community structure in very large networks. Physical Review E, 70(6), 066111.

Brin, S. and Page, L. (1998). The anatomy of a large-scale hypertextual web search engine. Computer Networks and ISDN Systems, 30(1-7), pp. 107-117.

Aroussi, R. (2017). yfinance: Yahoo! Finance market data downloader. Python package, <https://github.com/ranaroussi/yfinance>.

McKinney, W. (2010). Data structures for statistical computing in Python. Proceedings of the 9th Python in Science Conference, pp. 56-61.

Hunter, J.D. (2007). Matplotlib: A 2D graphics environment. Computing in Science & Engineering, 9(3), pp. 90-95.

Bird, S., Klein, E. and Loper, E. (2009). Natural Language Processing with Python. O'Reilly Media.

Kaggle (n.d.). 100K Nvidia Tweets dataset. Retrieved from <https://www.kaggle.com/datasets/soheiltehranipour/100k-nvidia-tweets?resource=download>